a A broad and a mechanism-specific channel are merged into one disease graph and embedded

Broad channel

Disease genes

OpenTargets
Orphanet, OMIM
GenCC
AZ BIKG literature

Disease symptoms

HPO annotations
OMIM

GO gene-function similarity
96% of pairs

HPA cell-type expression
91% of pairs

Pfam protein domains
94% of pairs

HPO phenotype
35% of pairs

Rank-mean fusion

each pair is ranked in every channel; ranks are averaged; missing channel ranks last; best-ranked pairs kept, at least 3 per disease

331,234 edges
13,020 diseases
(after merging)

MechDist graph

13,021 diseases
498,155 weighted edges after merging
(23,086 in both channels)

Benchmark pairs removed

10,052 pairs → 488,103 training edges

node2vec embedding

weighted, 32 dimensions
= DisAtlas embeddings

Pair decoder

MLP on the element-wise product, 5-seed ensemble; scores any disease pair

Mechanism-specific channel

Disease–mechanism evidence

BIKG literature	gene, pathway, process and phenotype links (text-mined)
OpenTargets	genetic association, somatic mutation, animal model
CTD	marker and mechanism genes
MetaBase	curated pathway maps
Autoantibodies	autoantibody target genes
Reactome	*pathway gene sets (disease genes by enrichment)*

Bipartite graph

diseases		*entities*
3,677	Mechanisms	357
3,839	Pathways	1,642
3,757	Reactome	1,377
1,245	literature	265
2,437	Biological processes	2,234

Specific edges

diseases scored by shared entities; link weight = source families + ln(1 + dated documents); 50 nearest per disease

190,007 edges
5,194 diseases
(after merging)

Disease–pathology evidence

BIKG pathology	anatomy, cell type and phenotype (text-mined)

Bipartite graph

diseases		*entities*
2,448	Anatomical sites	2,819
1,406	Cell types	464
2,249	Phenotype terms	1,088

BIKG = AstraZeneca Biological Insight Knowledge Graph;
CTD = Comparative Toxicogenomics Database

DisAtlas embeddings

no drug or clinical-trial information enters the graph

b Retrospective benchmark built from clinical-trial outcomes

Trial co-investigation (IxIDN)

Drug–disease trials

all trial records:
419,240 trials,
38,389 drugs,
1,593 diseases

removed in turn:
38 non-disease indications
2,266 drugs with no specific mechanism of action
52,056 trials of > 2 drugs

kept: 299,122 trials,
32,023 drugs,
1,437 diseases

IxIDN

trial graph: diseases linked when the same drug was trialled for both

1,379 diseases
43,112 links

Positive pairs (IxIDN⁺)

linked in IxIDN: 27,415 candidate pairs in the 985-disease evaluation universe; 4,302 with a matched negative were checked by a language model; 2,182 kept (1,285 strong, 897 plausible)

Negative pairs (IxIDN⁻)

≥ 3 IxIDN steps apart, no shared trial drug: 155,375 candidates; matched to a positive on trial-graph connectivity, organ system, clinical area, name and ontology similarity, gene coverage; 2,182 matched negatives

IxIDN-Matched benchmark

4,364 pairs
(2,182 IxIDN⁺, 2,182 IxIDN⁻)
938 diseases

disease-disjoint split by MONDO-ID hash: 461 tune, 477 test diseases

test split: 1,094 pairs with both diseases in the test half (491 IxIDN⁺, 603 IxIDN⁻)

metric: AUROC time-censored

Task 1 Disease–disease association

decoder scores pairs;
AUROC on IxIDN-Matched

Task 2 Disease clustering

13,021 diseases,
K = 55 clusters

benchmark pairs never enter the graph (checked automatically)

dated trial starts (scoring only)

c Simulated prospective test, year by year

Dated edges

each graph edge in (a) dated by its first evidence

Yearly graph snapshots

for each decision year y, 1996–2021: evidence up to the end of year y−1 only

Snapshot embeddings

DisAtlas retrained on each snapshot

Score each drug

candidates: diseases in that year's model not yet trialled with the drug (median 2,241 in the test pool); score = maximum similarity to the drug's earlier trial diseases

Rank its new trials

event = a drug's first trial in a new disease in year y; percentile of that disease among comparison candidates

Events

74,345 new drug–disease trials in 8,064 drugs; 35,793 trials in 6,496 drugs among test-pool diseases (not used to tune DisAtlas)

Comparison sets (co-primary)

candidates of the same prior trial activity (popularity), or of the same activity and similar weighted degree in that year's graph; groups need ≥ 20 candidates (adjacent groups merged)

Endpoints vs weighted Adamic–Adar

median over drugs of each drug's median event percentile; share of drugs with best new trial in top 10%; paired bootstrap over drugs, 2,000 replicates